

Deep Dive 1: What is Generative AI?

What you'll learn

By the end of this lesson, you'll be able to:

- Define generative AI and how it differs from other AI types
- Recognize the key characteristics and technological foundations of generative AI
- Identify major capabilities and limitations of current generative AI

Generative AI Fundamentals:

Video Link: [Deep Dive 1: What is Generative AI](#)

This video introduces the concept of generative AI, focusing on its ability to create new content rather than just analyzing what already exists. We walk through how large language models (LLMs) like Claude actually work and the technological journey that made them possible, from algorithmic breakthroughs like the transformer architecture to vast training datasets and powerful computing. We also explain how these systems learn through pre-training and fine-tuning and discuss concepts like context windows and emergent capabilities.

Capabilities & limitations

Video Link: [Capabilities & limitations](#)

This video examines what generative AI can and cannot do effectively at this point in time. We highlight generative AI's versatility across language tasks, ability to maintain conversational flow, and capacity to switch between diverse tasks without additional training. We also address limitations including knowledge cutoff dates, hallucinations (factually incorrect outputs), context window constraints, and reasoning challenges. We emphasize how the field is evolving rapidly and explain that the most effective applications bring together the complementary strengths of humans and AI working together.

Key takeaways

- Generative AI creates new content (text, images, code) rather than just analyzing existing data
- Modern systems like LLMs were made possible by three key developments:
 - Algorithmic and architectural breakthroughs (especially the transformer architecture)
 - Vast amounts of digital training data
 - Dramatic increases in computational power
- Generative AI learns through two stages: pre-training (analyzing patterns across billions of examples) and fine-tuning (learning to follow instructions and provide helpful responses)
- Current capabilities include versatility across tasks, conversational awareness, and the ability to connect with external tools
- Current limitations include knowledge cutoff dates, potential for hallucinations, context window constraints, and challenges with complex reasoning
- The most effective applications combine human and AI strengths, with humans providing critical thinking, judgment, creativity, and ethical oversight

Exercises

Reflection

Before moving on, take a moment to consider:

- How does understanding the technical foundations of generative AI (like training data and pre-training/fine-tuning) change how you think about working with these systems?
- What ethical considerations come to mind after learning about how these systems work and their current limitations?

What's next

In the next lesson, we'll take a closer look at the first of the 4D competencies: Delegation. You'll learn how to make strategic decisions about dividing work between yourself and AI based on understanding both your goals and AI capabilities. This foundation will help you thoughtfully determine when and how to bring AI into your creative and problem-solving processes.

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